

Research Question 8

Year Level & Extension Usage

Question

Is there a correlation between grade level and the amount of extensions used?

NEW RESULT

$$p = 0.091$$

```
          Df Sum Sq Mean Sq F value Pr(>F)
Year          3    4.27   1.4232    2.219 0.0911 .
Residuals    93   59.65   0.6414
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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$$(F(3, 93) = 2.22, p < 0.1)$$

Year(3) comes from (# of groups) - 1

Residuals(93) comes from (# of rows) - (# of groups)

$F = 2.22$ means that there is some difference between extension usage and years, *however*, the p-value (0.0911) shows that there is a 9% chance this happened by chance, so basically this means its not significant.

Raw Values Summary

OLD DATA

1. The Freshmen (Count: 18) had an mean of 0.89 extensions used.
2. The Sophomores (Count: 29) had an mean of 1.24 extensions used.
3. The Juniors (Count: 20) had an mean of 1.25 extensions used.
4. The Seniors (Count: 18) had an mean of 1.44 extensions used.

NEW DATA

1. The Freshmen (Count: 17) had an mean of 0.94 extensions used.
2. The Sophomores (Count: 35) had an mean of 1.20 extensions used.
3. The Juniors (Count: 23) had an mean of 1.44 extensions used.
4. The Seniors (Count: 22) had an mean of 1.55 extensions used.

	Year	n	Mean_Extensions	SE
1	Freshman	17	0.9412	0.2005615
2	Sophomore	35	1.2000	0.1408259
3	Junior	23	1.4348	0.1642577
4	Senior	22	1.5455	0.1574592

Statistical Note

Test used: One-way ANOVA

OLD Result: $p = 0.243$

Conclusion

OLD CONCLUSION: Though the data is not significant ($p > 0.05$), this still tells us something. Our data shows a positive correlation: as grade level increases, the amount of extensions used increases slightly. Our extensions policy is helpful for all students and is not necessarily a “safety net” or a “holding hand” for lowerclassmen students who just began college. Despite this, due to the insignificance proven, our extension policy can help all students alike without hindering performance or drastically changing performance.